

MAINE-SOROA, Niger

WMO: 610960

Lat: 13.233N

Lon: 11.983E

Elev: 338

StdP: 97.33

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.8	14.1	-8.2	2.0	24.8	-8.0	2.0	25.4	8.0	22.6	7.3	22.7	1.5	40	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.2	42.5	21.0	41.7	21.0	40.7	21.0	26.5	33.7	26.1	33.3	25.6	32.8	2.1	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.8	29.3	24.4	20.2	28.8	24.0	19.7	28.4	85.0	33.8	82.9	33.3	81.1	32.9	32.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	6.0	4.9	DB	10.3	44.3	0.8	0.7	9.7	44.8	9.2	45.2	8.7	45.6	8.1	46.1	(4)
			WB	3.9	28.6	0.7	1.6	3.4	29.7	2.9	30.6	2.5	31.5	2.0	32.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.9	22.4	25.6	29.9	33.4	34.3	32.9	30.0	28.2	29.8	30.7	27.1	23.1	(6)
(7)		DBStd	4.32	2.81	3.27	2.57	2.03	1.75	2.12	2.19	1.79	1.76	1.40	2.21	2.27	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6914	383	436	616	703	753	686	620	564	594	642	513	404	(10)
(11)		CDD18.3	3874	126	203	358	453	494	436	362	306	344	384	263	147	(11)
(12)		CDH23.3	52798	1419	2483	5093	7178	7892	6549	4629	3327	4199	5216	3232	1580	(12)
(13)		CDH26.7	31631	624	1347	3199	4919	5459	4229	2469	1474	2242	3145	1833	688	(13)
(14)	Wind	WSAvg	2.1	2.3	2.3	2.3	2.0	2.1	2.5	2.3	1.7	1.5	1.6	2.1	2.1	(14)
(15)	Precipitation	PrecAvg	350	0	0	0	2	8	31	109	149	53	10	0	0	(15)
(16)		PrecMax	650	0	0	5	41	44	91	225	438	156	87	5	5	(16)
(17)		PrecMin	165	0	0	0	0	0	0	30	0	2	0	0	0	(17)
(18)		PrecStd	114	0	0	1	7	11	24	51	80	33	17	1	1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.7	40.0	41.7	43.6	43.7	42.1	39.6	36.7	39.6	40.2	38.9	36.6	(19)
(20)			MCWB	16.7	17.9	19.1	20.5	21.5	22.8	23.9	25.2	23.4	21.1	18.9	17.5	(20)
(21)		2%	DB	34.5	38.1	40.6	42.6	42.7	40.9	37.9	35.0	38.0	39.3	37.5	34.2	(21)
(22)			MCWB	15.7	17.5	18.4	20.3	21.6	23.0	24.3	25.4	23.8	20.9	18.3	16.3	(22)
(23)		5%	DB	32.5	36.2	39.3	41.7	41.8	39.9	36.5	33.8	36.7	38.5	36.3	32.4	(23)
(24)			MCWB	14.8	16.6	17.9	20.2	21.5	23.3	24.4	25.4	24.4	21.0	17.6	15.2	(24)
(25)		10%	DB	30.2	34.0	37.8	40.6	40.8	38.6	35.1	32.6	35.3	37.4	34.9	30.7	(25)
(26)			MCWB	13.7	15.6	17.1	20.1	21.8	23.4	24.4	25.3	24.7	21.0	17.0	14.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.5	19.8	22.4	25.1	26.3	26.6	26.7	27.2	27.1	25.9	21.5	18.2	(27)
(28)			MCDWB	35.1	37.6	36.2	38.2	37.9	36.8	33.8	32.9	34.3	34.9	35.9	35.4	(28)
(29)		2%	WB	16.1	18.0	20.0	24.0	25.2	25.6	26.0	26.5	26.4	25.1	19.1	16.8	(29)
(30)			MCDWB	33.5	36.2	37.9	37.6	37.1	35.5	33.4	32.4	33.4	33.8	35.7	33.0	(30)
(31)		5%	WB	15.0	16.9	18.6	23.2	24.6	25.1	25.5	26.0	25.9	24.5	18.2	15.8	(31)
(32)			MCDWB	31.7	34.8	37.4	36.3	36.3	34.9	32.8	31.9	32.8	32.9	35.1	31.4	(32)
(33)		10%	WB	14.0	15.8	17.6	22.2	24.0	24.6	25.1	25.6	25.5	23.9	17.3	14.8	(33)
(34)			MCDWB	29.7	33.6	36.9	35.8	35.5	34.4	32.3	31.0	32.2	32.0	34.0	29.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.6	15.8	15.6	14.5	13.2	11.4	9.8	8.9	11.1	13.9	15.5	15.5	(35)
(36)			MCDBR	17.9	17.5	16.9	15.3	14.5	12.8	11.5	10.5	13.1	15.1	16.3	17.2	(36)
(37)			MCWBR	8.0	7.5	7.0	6.4	6.1	4.2	3.2	3.3	3.8	5.4	6.8	7.6	(37)
(38)		5% WB	MCDBR	17.6	17.0	16.1	12.9	12.5	11.1	10.0	9.5	11.2	12.8	16.0	16.5	(38)
(39)			MCWBR	8.1	7.7	7.1	5.7	5.1	3.9	3.4	3.7	4.0	4.5	7.2	7.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.541	0.602	0.649	0.749	0.726	0.704	0.667	0.619	0.624	0.664	0.583	0.580	(40)	
(41)		taud	1.725	1.593	1.514	1.390	1.444	1.534	1.654	1.756	1.727	1.583	1.639	1.648	(41)	
(42)		Ebn at Noon	756	730	710	641	643	649	677	720	718	674	713	707	(42)	
(43)		Edh at Noon	228	271	299	337	314	283	252	231	238	270	247	241	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.72	6.15	6.57	6.73	6.68	6.33	5.92	5.73	6.15	6.09	5.85	5.54	(44)	
(45)		RadStd	0.15	0.14	0.29	0.23	0.26	0.24	0.17	0.25	0.13	0.19	0.16	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page